Roshan Kern

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EDUCATION

Massachusetts Institute of Technology | Cambridge, MA — Sep 2026 - 2031 (expected)
Ph.D. in Biological Engineering
Accepted for 2025 cohort, deferred to 2026 for personal reasons

Case Western Reserve University, College of Arts and Sciences | Cleveland, OH — Sep 2021 - Dec 2024
B.S. in Systems Biology (Bioinformatics Subspecialty)
Minors in Applied Data Science, Computer Science, Business Management
GPA 4.0

EXPERIENCE

Telora Fellowship — Jan 2026 - July 2026
Telora Fellow
Mentor: Eliam Medina
- $50k and mentorship to start a company
- Identified problems by interviewing ~100 people per week to establish clear value loss in time, money, or accuracy
- Fractional COO for legal tech startup Hourglass (hourglass.law) with $250k ARR and 25% MoM growth

Massachusetts Institute of Technology | Cambridge, MA — Sep 2024 - June 2025 (Jan-June 2025 On-Site)
Research Intern (Massachusetts Life Sciences Center Data Science Internship Program)
Advisor: Iain Cheeseman
- Co-founded computational team to make the lab cross-functional, implementing policies for project management, code review, and computational workflows.
- Co-developed Brieflow for processing optical pooled screen data at scale with a focus on modularity/extensibility for effective process development.
- Cut screen configuration time by 80% (5→1 day) and module integration time by 93% (2 weeks→1 day)

Recursion Pharmaceuticals | Salt Lake City, UT — May 2024 - Aug 2024
Data Science Intern
Mentor: Kyle Hansen
- Established metric for and optimized gene target selection from large-scale gene-chemical interactions.
- Investigated contribution of embeddings from three phenomic maps to compound selection process.

Case Western Reserve University | Cleveland, OH — May 2023 - Aug 2023
Summer Undergraduate Research Program Fellow
Advisor: Thomas LaFramboise
- Implemented pipeline for human-endogenous retrovirus profiling from patient transcriptomics data.
- Investigated effect of human-endogenous retrovirus expression on bone marrow failure syndromes.

University of Colorado Anschutz | Aurora, CO — May 2022 - May 2023
Research Intern
Advisor: Gregory Way
- Trained machine learning model for classification of single-cell images into one of 15 phenotypes using morphology features.

- Achieved F1 score of 0.84 on held-out test set and applied model to different cell image datasets to understand generalizability.
- Developed cell image processing tool for extracting features from single-cell images (IDR Stream).

PUBLICATIONS

5. Marescal, O., Su, K.-C., Di Bernardo, M., Manzer, K., He, J., **Kern, R.**, Cheeseman, I.M. (2026) Functional genetics reveals a role for the proteasome in regulating centromere assembly. ***Current Biology*** (in review)

4. Nguyen, A.D.D., **Kern, R.**, Dombrovski, T., LaFramboise, T. (2026) The circulating metatranscriptome in acute myeloid leukemia patients: an RNA-seq reanalysis. ***Frontiers in Microbiomes*** 5:1903806: doi.org/10.3389/frmbi.2026.1903806

3. Di Bernardo, M., **Kern, R.**, Cepeda Diaz, A.K., Mallar, A., Choi, S.J., Nutter-Upham, A., Lourido, S., Blainey, P.C., Cheeseman, I.M. (2026) Brieflow: an integrated computational pipeline for high-throughput analysis of optical pooled screening data. ***Nature Communications***: doi.org/10.1038/s41467-026-73643-7

2. Serrano, E., Chandrasekaran, S.N., Bunten, D., Brewer, K.I., Tomkinson, J., **Kern, R.**, Bornholdt, M., Fleming, S., Pei, R., Arevalo, J., Tsang, H., Rubinetti, V., Tromans-Coia, C., Becker, T., Weisbart, E., Bunne, C., Kalinin, A.A., Senft, R., Taylor, S.J., Jamali, N., Adeboye, A., Abbasi, H.S., Goodman, A., Caicedo, J.C., Carpenter, A.E., Cimini, B.A., Singh, S., Way, G.P. (2025) Reproducible image-based profiling with Pycytominer. ***Nature Methods***: doi.org/10.1038/s41592-025-02611-8

1. Tomkinson, J.*, **Kern, R.***, Mattson, C., Way, G.P. (2024) Toward generalizable phenotype prediction from single-cell morphology representations. ***BMC Methods***: doi.org/10.1186/s44330-024-00014-3, **Co-first authors*

MAJOR PRODUCTS

Brieflow: Open source, extensible tool for processing data from Optical Pooled Screens. (github.com/cheeseman-lab/brieflow, anthropic.com/news/accelerating-scientific-research)

Pycytominer: Open source suite of common functions used to process high dimensional readouts from high-throughput cell experiments. (github.com/cytomining/pycytominer)

IDR Stream: Open source software to handle cell image downloading, preprocessing, cell segmentation, feature extraction, feature compilation for Image Data Resource datasets. (github.com/WayScience/IDR_stream)

HONORS AND FELLOWSHIPS

2026	MIT Prize for Open Data (Brieflow)
2026	MIT Presidential Fellowship
2026	Telora Fellow
2024	Case Alumni Association Junior Senior Scholar
2024	Dean's High Honors List, Case Western Reserve University
2023	Cancer-focused Summer Undergraduate Research Fellow, Case Western Reserve University

PRESENTATIONS AND TEACHING

CWRU CanSUR Symposium: "Human Endogenous Retrovirus Landscape of Aplastic Anemia", *August 2023*

Way Lab: Led lab discussions of self-supervised machine learning for live cell imagery segmentation (Robitaille et al., 2022) and the development of a dynamic protein atlas of human cell division (Cai et al., 2018).

Recursion Pharmaceuticals: Presented work on gene and compound selection processes to senior data scientists at data analysis reviews.

COMMUNITY

Nucleate Boston Spring 2025
Co-Director of Strategy and Operations
- Organized educational workshop series for early-stage biotech founders in an equity-free accelerator.
- Recruited industry mentors and sponsors for the series.

Campus Flavor | Cleveland, OH Jan 2023 - May 2023
Project Leader/Coordinator
- Led two technical developers to create a website for collecting university-specific merchandise designs.
- Received 32 student-created designs in two weeks of hosting competition on the website.
- Communicated with business representatives from API company and university to formulate business strategy.

Saturday Tutoring Program | Cleveland, OH Jan 2023 - May 2023
Middle School Tutor
- Tutored middle school students in STEM topics including biology, chemistry, and physics.

Animal Protective League | Cleveland, OH Sep 2023 - Dec 2023
Dog Walker
- Walked and played with dogs at APL animal shelter.

MedWish | Cleveland, OH Jan 2022 - May 2023
Member
- Organized and inventoried donated medical supplies for distribution to areas in need.
- Led eight novice programmers in building a club website to attract new members.